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CUET UG

Chemistry

# Chemical Kinetics

Official Syllabus Unit III · Class 12 · CUET UG 2027 Cycle Notes

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# Syllabus Alignment

| Sub-topic | Official Syllabus Pointer (NTA CUET UG Baseline) | Skills & Competencies Tested |
|---|---|---|
| **Rate of Reaction & Rate Law** | Rate of a reaction (average and instantaneous), factors affecting rate: concentration, temperature, catalyst; order and molecularity of a reaction. | Formulating stoichiometric rate expressions, determining order from rate laws or experimental initial-rate data tables, and contrasting order with molecularity. |
| **Integrated Rate Equations & Half-Life** | Integrated rate equations and half-life for zero and first order reactions; pseudo-first-order reactions; gaseous phase first order decomposition. | Applying $[R] = [R]_0 - kt$ and $k = \left(\frac{2.303}{t}\right)\log\left(\frac{[R]_0}{[R]}\right)$, computing $t_{\frac{1}{2}}$, solving multi-step fractional decay $(t_{75\%}, t_{99.9\%})$, and analyzing partial pressures. |
| **Temperature Dependence & Arrhenius Equation** | Concept of collision theory (elementary ideas, no mathematical treatment), activation energy, Arrhenius equation, catalysis energetics. | Applying $\log\left(\frac{k_2}{k_1}\right) = \left(\frac{E_a}{2.303}R\right)\left[\frac{T_2 - T_1}{T_1 T_2}\right]$, reading slopes of $\log k$ vs $\frac{1}{T}$, and evaluating the effect of catalysts on $E_a, \Delta H$, and $K_c$. |

**EXAM PATTERN**

**CUET UG Exam Strategy:** Unit III Chemical Kinetics regularly yields 4 to 5 questions in the CUET UG paper (+5, −1 mark scheme). The highest-yield concepts tested include: (1) reading reaction order from the physical units of $k$ ($s^{-1}$ for 1st order, mol $L^{-1}$ $s^{-1}$ for zero order, L $mol^{-1}$ $s^{-1}$ for 2nd order), (2) the independence of 1st order half-life on initial concentration ($t_{\frac{1}{2}} = \frac{0.693}{k}$), (3) multi-step relations like $t_{75\%} = 2 \times t_{50\%}$ and $t_{99.9\%} \approx 10 \times t_{50\%}$, and (4) the fact that a catalyst lowers $E_a$ but leaves $\Delta H$ and equilibrium constant $K_c$ completely unchanged.

# Concept Module 1: Rate of Chemical Reaction

**Chemical Kinetics** investigates the speeds of chemical reactions, the factors governing their rates, and the molecular mechanisms by which they proceed.

- **Thermodynamics vs Kinetics:** Thermodynamics predicts whether a reaction is feasible ($\Delta G < 0$); kinetics determines **how fast** that reaction occurs.
- **Classification by Speed:**
  - **Instantaneous Reactions:** Ionic reactions like precipitation of $AgCl$ ($AgNO_3 + NaCl \rightarrow AgCl\downarrow + NaNO_3$) occur within $10^{-12}$ to $10^{-16}$ seconds.
  - **Extremely Slow Reactions:** Rusting of iron, weathering of rocks require months or years.
  - **Moderate (Measurable) Reactions:** Molecular reactions like inversion of cane sugar, hydrolysis of starch, or decomposition of $N_2O_5$ proceed at measurable rates under ambient conditions.

## Average Rate vs Instantaneous Rate

Consider a general reaction: $R \rightarrow P$

1. **Average Rate ($r_{\text{av}}$):** The change in molar concentration of reactants or products divided by the finite time interval ($\Delta t$):

$$r_{\text{av}} = -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t}$$

   The negative sign denotes that reactant concentration **decreases** over time.

2. **Instantaneous Rate ($r_{\text{inst}}$):** The rate of reaction at a specific infinitesimal instant of time ($\Delta t \rightarrow 0$), determined from the slope of the tangent to the concentration-time curve:

$$r_{\text{inst}} = -\frac{d[R]}{dt} = +\frac{d[P]}{dt}$$

## Stoichiometric Rate Expressions

For a reaction with arbitrary stoichiometric coefficients:

$$aA + bB \rightarrow cC + dD$$

The rate of reaction is defined by dividing the individual rates of disappearance or appearance by their stoichiometric coefficients:

$$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt}$$

- **Units of Reaction Rate:** Always mol $L^{-1}$ $s^{-1}$ (or mol $L^{-1}$ $min^{-1}$). For gaseous reactions: atm $s^{-1}$ or bar $s^{-1}$.

**WORKED EXAMPLE 1 – STOICHIOMETRIC RATE RELATIONS**

In the decomposition of dinitrogen pentoxide:

$$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$$

If the rate of formation of $NO_2$ is 0.0072 mol $L^{-1}$ $s^{-1}$, calculate: (a) the rate of reaction, and (b) the rate of disappearance of $N_2O_5$.

**Solution:**

$$\text{Rate of reaction} = +\frac{1}{4}\frac{d[NO_2]}{dt} = \frac{1}{4} \times 0.0072 = 0.0018 \text{ mol } L^{-1} \text{ } s^{-1}$$

$$\text{Rate of disappearance of } N_2O_5 = -\frac{d[N_2O_5]}{dt} = 2 \times \text{Rate} = 2 \times 0.0018 = 0.0036 \text{ mol } L^{-1} \text{ } s^{-1}$$

## Factors Influencing Rates of Chemical Reactions

1. **Reactant Concentration:** Elevates collision frequency per unit volume, directly scaling rate according to the differential rate law.
2. **Surface Area in Heterogeneous Systems:** Subdividing solid reactants into fine powders exposes more active surface sites, dramatically accelerating reaction velocity.
3. **Temperature:** A $10^{\circ}$ C rise typically doubles or triples reaction rates by exponentially increasing the molecular fraction exceeding activation energy $E_a$.
4. **Catalyst Action:** Provides an alternate mechanism with lowered $E_a$, accelerating both forward and reverse rates without affecting equilibrium position.
5. **Photochemical Activation:** Light photons ($h\nu$) supply activation energy in specific reactions (e.g., $H_2 + Cl_2 \xrightarrow{h\nu} 2\ HCl$).

# Concept Module 2: Rate Law, Order & Molecularity

## The Rate Law Formulation

The **Rate Law** is the mathematical expression in which reaction rate is expressed in terms of the molar concentrations of reactants with each term raised to some power:

$$\text{Rate} = k[A]^x[B]^y$$

where $k$ is the **Rate Constant** (or **Specific Reaction Rate**).

- **Critical Distinction:** The exponents $x$ and $y$ are **determined solely by experiment** and may or may not equal the stoichiometric coefficients $a$ and $b$ of the balanced chemical equation.

## Order of a Reaction

The **Overall Order ($n$)** is the sum of the powers of the concentration terms in the experimental rate law:

$$n = x + y$$

- A reaction can have an order of zero, integer $(1, 2, 3)$, fraction $(\frac{1}{2}, \frac{3}{2})$, or even negative.
- **Units of Rate Constant ($k$):**

$$\text{Units of } k = (\text{mol L}^{-1})^{1-n} \times \text{s}^{-1}$$

  - **Zero Order ($n = 0$):** mol $L^{-1}$ $s^{-1}$ (same as rate units).
  - **First Order ($n = 1$):** $s^{-1}$ or $min^{-1}$ (independent of concentration units).
  - **Second Order ($n = 2$):** L $mol^{-1}$ $s^{-1}$ (or $mol^{-1}$ L $s^{-1}$).
  - **Third Order ($n = 3$):** $L^2$ $mol^{-2}$ $s^{-1}$.
  - **Half Order ($n = \frac{1}{2}$):** $mol^{\frac{1}{2}}$ $L^{-\frac{1}{2}}$ $s^{-1}$.

## Determination of Rate Law from Initial Rate Data

In laboratory kinetics, the method of initial rates measures the initial reaction rate over several runs while varying one reactant concentration at a time:

### WORKED EXAMPLE 2 — INITIAL RATES METHOD ANALYSIS

Consider the reaction $2A + B \rightarrow C + D$. The initial rate was measured at several different concentrations:

| $[A]$ (mol/L) | $[B]$ (mol/L) | Initial Rate (mol/L·s) |
|---|---|---|
| 0.10 | 0.10 | $2.0 \times 10^{-3}$ |
| 0.20 | 0.10 | $8.0 \times 10^{-3}$ |

| $[A]$ (mol/L) | $[B]$ (mol/L) | Initial Rate (mol/L·s) |
|---|---|---|
| 0.10 | 0.20 | $4.0 \times 10^{-3}$ |

**Determine:** (a) Order with respect to $A$, (b) Order with respect to $B$, (c) Rate law, and (d) Rate constant $k$.

**Solution:** Let Rate $= k[A]^x[B]^y$.

- Comparing Exp 1 & 2 (with $[B]$ held constant at 0.10):

$$\frac{\text{Rate}_2}{\text{Rate}_1} = \frac{8.0 \times 10^{-3}}{2.0 \times 10^{-3}} = \left(\frac{0.20}{0.10}\right)^x \implies 4 = 2^x \implies x = 2$$

- Comparing Exp 1 & 3 (with $[A]$ held constant at 0.10):

$$\frac{\text{Rate}_3}{\text{Rate}_1} = \frac{4.0 \times 10^{-3}}{2.0 \times 10^{-3}} = \left(\frac{0.20}{0.10}\right)^y \implies 2 = 2^y \implies y = 1$$

- Rate Law: $\mathbf{Rate = k[A]^2[B]}$ (Overall order $= 2 + 1 = 3$).
- Solving for $k$ using Exp 1:

$$k = \frac{\text{Rate}}{[A]^2[B]} = \frac{2.0 \times 10^{-3}}{(0.10)^2 \times 0.10} = \frac{2.0 \times 10^{-3}}{1.0 \times 10^{-3}} = 2.0 \text{ L}^2 \text{ mol}^{-2} \text{ s}^{-1}$$

## Molecularity of an Elementary Reaction

The **Molecularity** is the number of reacting species (atoms, ions, or molecules) taking part in an elementary step which must collide simultaneously to bring about chemical change.

- **Unimolecular (1):** $NH_4NO_2 \rightarrow N_2 + 2H_2O$
- **Bimolecular (2):** $2HI \rightarrow H_2 + I_2$
- **Trimolecular (3):** $2NO + O_2 \rightarrow 2NO_2$ (Reactions with molecularity $> 3$ are extremely rare because simultaneous collision of 4 or more particles is statistically improbable).

| Comparison Parameter | Order of Reaction | Molecularity of Reaction |
|---|---|---|
| **Definition** | Sum of concentration exponents in experimental rate law. | Number of species colliding in an elementary step. |
| **Nature** | Strictly experimental quantity; cannot be deduced from equation. | Theoretical concept deduced from reaction mechanism. |
| **Allowable Values** | Can be zero, whole number, fraction, or negative. | Only positive whole integers (1, 2, or 3); never zero or fraction. |
| **Scope** | Applicable to both elementary and complex reactions. | Meaningful **only** for elementary single-step reactions. |
| **Dependence** | Can change with experimental temperature or pressure. | Invariant; fixed solely by the reaction mechanism. |

## Complex Multi-Step Reactions & The Rate Determining Step (RDS)

Most chemical transformations occur via consecutive elementary steps (**reaction mechanism**):

- **The Rate Determining Step (RDS):** The slowest elementary step acts as the kinetic bottleneck, dictating the overall reaction velocity.
- **Thermal Decomposition of $N_2O_5$ ($2N_2O_5 \rightarrow 4NO_2 + O_2$):** Proceeds via: (1) $N_2O_5 \leftrightarrow NO_2 + NO_3$ (fast equilibrium); (2) $NO_2 + NO_3 \rightarrow NO + NO_2 + O_2$ (slow, RDS); (3) $NO + NO_3 \rightarrow 2NO_2$ (fast). Substituting the equilibrium concentration of intermediate gives the experimental first-order rate law: Rate $= k[N_2O_5]$.

- **Anomalous Negative Temperature Coefficient:** Oxidation of nitric oxide ($2\ \mathrm{N\ O} + \mathrm{O_2} \rightarrow 2\mathrm{N\ O_2}$) slows down upon heating because a rapid exothermic pre-equilibrium ($2\ \mathrm{N\ O} \leftrightarrow \mathrm{N_2O_2}, \Delta H < 0$) shifts backward at higher temperatures faster than the rate constant increases.

# Concept Module 3: Integrated Rate Equations & Half-Life

## 1. Zero Order Reactions

A reaction where the rate is proportional to the zero power of the reactant concentration:

$$-\frac{d[R]}{dt} = k[R]^0 = k \Longrightarrow -d[R] = kdt$$

Integrating from $t = 0$ ($[R] = [R]_0$) to time $t$ ($[R]$):

$$[R] = [R]_0 - kt \quad \text{or} \quad k = \frac{[R]_0 - [R]}{t}$$

- **Linear Plot:** Plot of $[R]$ against $t$ is a straight line with:
  - $y$-intercept $= [R]_0$
  - Slope $= -k$
- **Half-Life ($t_{(\frac{1}{2})}$):** The time required for reactant concentration to reduce to half its initial value ($[R] = \frac{[R]_0}{2}$):

$$t_{\frac{1}{2}} = \frac{[R]_0}{2k}$$

  **Key Rule:** For zero order, $t_{\frac{1}{2}}$ is **directly proportional** to initial concentration $[R]_0$.
- **Time for 100% Completion:** $t_{100\%} = \frac{[R]_0}{k} = 2 \times t_{\frac{1}{2}}$.

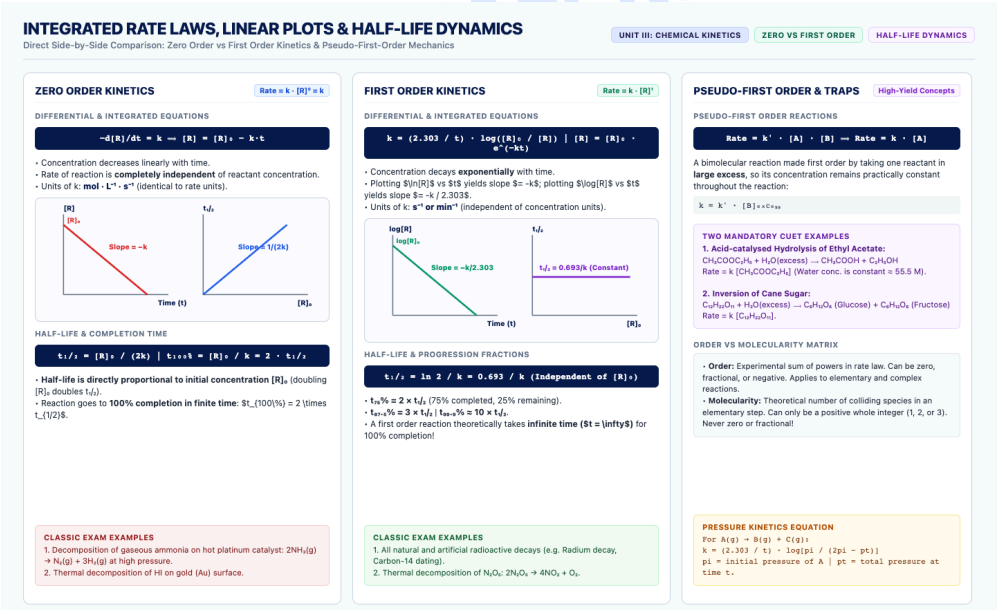



Figure 1: Integrated Rate Laws: Zero Order vs First Order linear kinetics, half-life concentration invariance, and pseudo-first-order solvent excess mechanics.

## 2. First Order Reactions

A reaction where the rate is directly proportional to the first power of reactant concentration:

$$-\frac{d[R]}{dt} = k[R] \Longrightarrow \frac{d[R]}{[R]} = -kdt$$

Integrating: $\ln[R] = -kt + \ln[R]_0 \Longrightarrow [R] = [R]_0 e^{-kt}$ Converting to common logarithm (base 10):

**FIRST ORDER INTEGRATED RATE EQUATIONS**

$$k = \frac{2.303}{t}\log\left(\frac{[R]_0}{[R]}\right) = \frac{2.303}{t}\log\left(\frac{a}{a-x}\right)$$

$$\log[R] = \log[R]_0 - \left(\frac{k}{2.303}\right)t$$

- Plotting $\log[R]$ vs $t$ yields a straight line with $y$-intercept $= \log[R]_0$ and **Slope** $= -\frac{k}{2.303}$.

## Half-Life of a First Order Reaction ($t_{(\frac{1}{2})}$):

Substitute $[R] = \frac{[R]_0}{2}$ at $t = t_{\frac{1}{2}}$:

$$t_{\frac{1}{2}} = \frac{2.303}{k}\log\left(\frac{[R]_0}{\frac{[R]_0}{2}}\right) = \frac{2.303}{k}\log 2 = \frac{2.303 \times 0.3010}{k}$$

$$t_{\frac{1}{2}} = \frac{0.693}{k}$$

**FUNDAMENTAL FIRST ORDER HALF-LIFE RULES**

1. **Concentration Independence:** $t_{\frac{1}{2}}$ is **completely independent** of initial concentration $[R]_0$. Doubling or tripling $[R]_0$ has zero impact on the time required to consume 50% of the reactant.
2. **Progressive Multi-Half-Life Relations:**
   - $t_{75\%} = 2 \times t_{50\%}$ (75% completed $\rightarrow$ 25% remaining $= \left(\frac{1}{2}\right)^2$).
   - $t_{87.5\%} = 3 \times t_{50\%}$ (12.5% remaining $= \left(\frac{1}{2}\right)^3$).
   - $t_{99.9\%} \approx 10 \times t_{50\%}$ ($[R] = \frac{[R]_0}{1000}$).
3. **Radioactive Decay:** All natural and artificial radioactive disintegrations follow **first-order kinetics** ($N = N_0 e^{-\lambda t}, t_{\frac{1}{2}} = \frac{0.693}{\lambda}$).

**WORKED EXAMPLE 3 — FIRST ORDER HALF-LIFE PROOFS**

**Prove that for a first order reaction, the time required for 99.9% completion is approximately 10 times the half-life ($t_{(\frac{1}{2})}$).**

**Solution:**

1. For half-life (50% completion):

$$t_{50\%} = \frac{2.303}{k}\log\left(\frac{[R]_0}{0.50[R]_0}\right) = \frac{2.303}{k}\log 2 = \frac{0.693}{k}$$

2. For 99.9% completion ($[R] = [R]_0 - 0.999[R]_0 = 0.001[R]_0$):

$$t_{99.9\%} = \frac{2.303}{k}\log\left(\frac{[R]_0}{0.001[R]_0}\right) = \frac{2.303}{k}\log 10^3 = \frac{2.303 \times 3}{k} = \frac{6.909}{k}$$

3. Comparing the two times:

$$\frac{t_{99.9\%}}{t_{50\%}} = \frac{\frac{6.909}{k}}{\frac{0.693}{k}} = 9.97 \approx 10 \Longrightarrow t_{99.9\%} \approx 10 \times t_{\frac{1}{2}}$$

## Comprehensive Kinetics Comparison Across Reaction Orders

| Kinetic Feature | Zero Order ($n = 0$) | First Order ($n = 1$) | Second Order ($n = 2$) |
|---|---|---|---|
| Differential Rate | $-\frac{d[R]}{d}t = k$ | $-\frac{d[R]}{d}t = k[R]$ | $-\frac{d[R]}{d}t = k[R]^2$ |
| Integrated Form | $[R] = [R]_0 - kt$ | $\ln\left(\frac{[R]_0}{[R]}\right) = kt$ | $\frac{1}{[R]} - \frac{1}{[R]_0} = kt$ |

| Kinetic Feature | Zero Order ($n = 0$) | First Order ($n = 1$) | Second Order ($n = 2$) |
|---|---|---|---|
| Linear Plot | $[R]$ vs $t$ (Slope $= -k$) | $\log[R]$ vs $t$ (Slope $= -\frac{k}{2.303}$) | $\frac{1}{[R]}$ vs $t$ (Slope $= +k$) |
| Half-Life ($t_{\frac{1}{2}}$) | $t_{\frac{1}{2}} = \frac{[R]_0}{2k}$ ($\propto [R]_0$) | $t_{\frac{1}{2}} = \frac{0.693}{k}$ ($\propto [R]_0^0$) | $t_{\frac{1}{2}} = \frac{1}{k[R]_0}$ ($\propto \frac{1}{[R]_0}$) |
| 100% Completion | $t_{100\%} = \frac{[R]_0}{k} = 2t_{\frac{1}{2}}$ | $t_{100\%} = \infty$ (Infinite time) | $t_{100\%} = \infty$ (Infinite time) |
| Units of $k$ | mol $L^{-1}$ $s^{-1}$ | $s^{-1}$ | L $mol^{-1}$ $s^{-1}$ |

# Concept Module 4: Gaseous Kinetics & Pseudo-First-Order

## Integrated Rate Equation for Gaseous Phase Reactions

For a typical first-order gas-phase decomposition:

$$A(g) \rightarrow B(g) + C(g)$$

- Let initial pressure of $A$ at $t = 0$ be $p_i$.
- At time $t$, let $x$ atm of $A$ decompose to form $x$ atm of $B$ and $x$ atm of $C$:
  - Partial pressure of $A$: $p_A = p_i - x$
  - Partial pressure of $B$: $p_B = x$
  - Partial pressure of $C$: $p_C = x$
- Total pressure at time $t$ ($p_t$):

$$p_t = p_A + p_B + p_C = (p_i - x) + x + x = p_i + x \Longrightarrow x = p_t - p_i$$

- Substituting $x$ back into $p_A$:

$$p_A = p_i - (p_t - p_i) = 2p_i - p_t$$

- Applying the first-order rate equation:

$$k = \frac{2.303}{t} \log\left(\frac{p_i}{p_A}\right) = \frac{2.303}{t} \log\left(\frac{p_i}{2p_i - p_t}\right)$$

## Pseudo-First-Order Reactions

A reaction that is truly of higher molecularity (e.g. bimolecular) but behaves kinetically as **first order** because one of the reacting species is present in **huge stoichiometric excess**.

- Because the concentration of the excess reactant changes negligibly during the entire reaction, its concentration is absorbed into an effective pseudo-first-order rate constant $k$:

$$\text{Rate} = k'[A][B]_{\text{excess}} \Longrightarrow \text{Rate} = k[A] \quad (\text{where } \; k = k'[B]_{\text{excess}})$$

### Classic CUET UG Examples of Pseudo-First-Order Reactions

1. **Acid-Catalysed Hydrolysis of Ethyl Acetate:**

$$CH_3COOC_2H_5 + H_2O(\text{excess}) \xrightarrow{H^+} CH_3COOH + C_2H_5OH$$

Water is present at $\approx 55.5$ M, while ester concentration is $\approx 0.01$ M. Even when all ester reacts, water concentration changes from 55.5 M to 55.49 M, remaining effectively constant.

$$\text{Rate} = k[CH_3COOC_2H_5]$$

2. **Inversion of Cane Sugar (Sucrose):**

$$C_{12}H_{22}O_{11} + H_2O(\text{excess}) \xrightarrow{H^+} C_6H_{12}O_6(\text{D-glucose}) + C_6H_{12}O_6(\text{D-fructose})$$

- **Polarimeter Observation:** Sucrose is dextrorotatory ($+66.5^0$). Upon hydrolysis, equimolar D-glucose ($+52.5^0$) and D-fructose ($-92.4^0$) form. Because fructose's laevorotation exceeds glucose's dextrorotation, the overall optical rotation flips from dextro ($+$) to laevo ($-$), hence termed **inversion of sugar**.

$$k = \frac{2.303}{t} \log \frac{r_0 - r_\infty}{r_t - r_\infty}$$

where $r_0, r_t$, and $r_\infty$ are optical angle readings at $t = 0, t = t$, and $t = \infty$.

# Concept Module 5: Temperature Dependence & Arrhenius Theory

For almost all chemical reactions, an increase in temperature markedly increases the reaction rate.

- **The Temperature Coefficient:** For every $10^0$C rise in temperature, the rate constant doubles or triples:

$$\text{Temperature Coefficient} = \frac{k_{T+10}}{k_T} \approx 2 \text{ to } 3$$

## The Arrhenius Equation

Svante Arrhenius proposed the quantitative formulation:

$$k = Ae^{-\frac{E_a}{RT}}$$

- $k$: Rate constant.
- $A$: **Arrhenius Pre-Exponential Factor** (Frequency Factor), representing the collision frequency of molecules.
- $E_a$: **Activation Energy** (in J mol$^{-1}$ or kJ mol$^{-1}$).
- $R$: Universal gas constant (8.314 J K$^{-1}$mol$^{-1}$).
- $T$: Absolute temperature in Kelvin.
- $e^{-\frac{E_a}{RT}}$: The **Boltzmann Fraction** of molecules possessing kinetic energy greater than or equal to $E_a$.

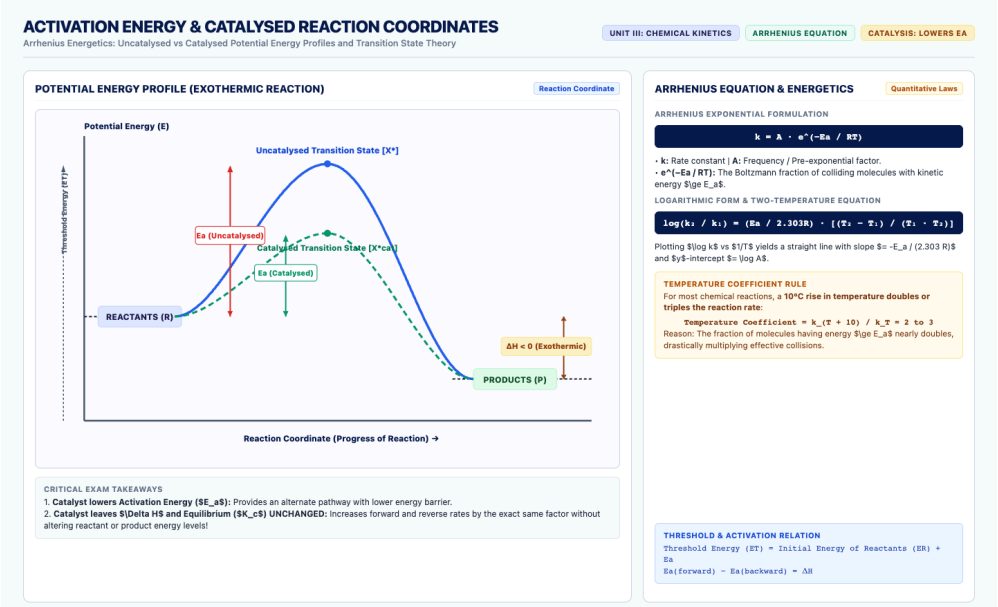



Figure 2: Potential Energy Profile: Activation energy barrier ($E_a$), transition state (X*), and the lowering of $E_a$ via an alternate catalysed reaction pathway.

## Logarithmic Forms of Arrhenius Equation

Taking natural logarithm (ln):

$$\ln k = \ln A - \frac{E_a}{RT}$$

Converting to common base-10 logarithm:

$$\log k = \log A - \frac{E_a}{2.303RT}$$

- **Linear Arrhenius Plot:** Plotting $\log k$ against $\frac{1}{T}$ yields a straight line:
  - $y$-intercept $= \log A$

- **Slope** $= -\frac{E_a}{2.303R}$
- $E_a = -2.303R \times \text{Slope}$

## Calculation of $E_a$ at Two Different Temperatures

If rate constants are $k_1$ and $k_2$ at absolute temperatures $T_1$ and $T_2$:

**TWO-TEMPERATURE ARRHENIUS EQUATION**

$$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 \times T_2}\right]$$

**WORKED EXAMPLE 4 — ARRHENIUS ACTIVATION ENERGY CALCULATION**

The rate constant of a reaction doubles when the temperature increases from 298 K to 308 K. Calculate the activation energy ($E_a$) of the reaction. ($R = 8.314$ J K$^{-1}$mol$^{-1}$, $\log 2 = 0.3010$).

**Solution:** Given $\frac{k_2}{k_1} = 2, T_1 = 298$ K, $T_2 = 308$ K.

$$\log 2 = \frac{E_a}{2.303 \times 8.314}\left[\frac{308 - 298}{298 \times 308}\right]$$

$$0.3010 = \frac{E_a}{19.147}\left[\frac{10}{91784}\right] \Longrightarrow E_a = \frac{0.3010 \times 19.147 \times 91784}{10}$$

$$E_a = 52897 \text{ J mol}^{-1} = 52.9 \text{ kJ mol}^{-1}$$

## Maxwell-Boltzmann Energy Distribution & The 10 K Rate Doubling Rule

At temperature $T$, molecules possess a distribution of kinetic energies with peak at the most probable kinetic energy ($E_{\text{mp}}$). When temperature rises to $T + 10$:

1. The curve flattens and broadens, shifting toward higher energies.
2. The fraction of molecules possessing energy $E \geq E_a$ (the shaded area beyond $E_a$) doubles or triples:

$$x = \frac{N_E}{N_T} = e^{-\frac{E_a}{RT}}$$

Taking logarithms: $\log x = -\frac{E_a}{2.303RT}$, confirming the exponential sensitivity of reaction rate to temperature.

# Concept Module 6: Collision Theory & Mechanism of Catalysis

## Collision Theory of Chemical Reactions

Developed by Max Trautz and William Lewis, based on the kinetic molecular theory of gases.

- **Fundamental Postulate:** Reactant molecules are hard spheres; reactions occur when molecules collide with one another.
- **Rate of Reaction Formula:**

$$\text{Rate} = P \times Z_{AB} \times e^{-\frac{E_a}{RT}}$$

  where $Z_{AB}$ is the **collision frequency** (number of collisions per second per unit volume), and $P$ is the **Steric Factor** (Orientation Factor).
- **Two Criteria for an Effective Collision:**
  1. **Energy Barrier:** Colliding molecules must possess a minimum energy termed the **Threshold Energy ($E_T$)**:

$$E_T = \text{Initial Energy of Reactants} + E_a$$

2. **Orientation Barrier:** Reacting molecules must collide with proper spatial orientation so that reactive chemical bonds can break and new bonds can form.

## Catalysis & Its Impact on Chemical Energetics

A **catalyst** is a substance that alters the rate of a chemical reaction without undergoing any permanent chemical change itself.

- **Action Mechanism:** An intermediate complex forms between reactants and catalyst, providing an **alternate reaction pathway with a lower activation energy barrier ($E_a$)**.
- **Key Exam Facts Regarding Catalysts:**
  1. A catalyst **lowers $E_a$** equally for both forward and reverse reactions.
  2. A catalyst **increases reaction rate** equally in both directions, accelerating the attainment of equilibrium.
  3. A catalyst **DOES NOT alter:**
     - The enthalpy change of reaction ($\Delta_r H$).
     - The Gibbs free energy of reaction ($\Delta_r G$).
     - The equilibrium constant ($K_c$).
     - The equilibrium concentrations of reactants and products.

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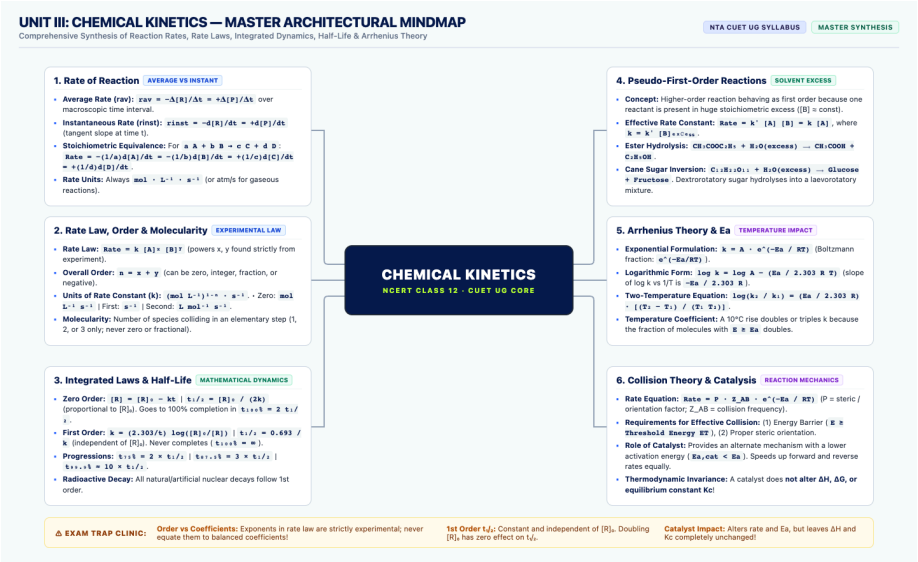



Figure 3: Unit III Chemical Kinetics Master Mindmap: Comprehensive synthesis of rates, order vs molecularity, integrated rate laws, temperature scaling, Arrhenius kinetics, and collision theory.

## MASTER KINETICS ARCHITECTURE & REACTION DYNAMICS BREAKDOWN

The cognitive map above consolidates the mathematical foundations and core concepts across Unit III:

- **Rate Law Formulation:** Always differentiate between rate units (mol $L^{-1}$ $s^{-1}$ universally) and rate constant units ($(\text{mol } L^{-1})^{1-n}$ $s^{-1}$). Order $n$ can be fractional or zero; molecularity is strictly a positive integer $(1, 2, 3)$.
- **Integrated Kinetics & Half-Life Multipliers:** For zero order, $t_{\frac{1}{2}} \propto [R]_0$ and $t_{100\%} = 2 \times t_{\frac{1}{2}}$. For first order, $t_{\frac{1}{2}}$ is concentration-independent ($\frac{0.693}{k}$), yielding fixed milestones: $t_{75\%} = 2t_{50\%}, t_{87.5\%} = 3t_{50\%}$, and $t_{99.9\%} \approx 10t_{50\%}$.
- **Gaseous Reactions Pressure Kinetics:** In gas-phase decompositions $A(g) \rightarrow B(g) + C(g)$, remember that partial pressure $p_A = 2p_i - p_t$. Never substitute raw total pressure directly into the logarithmic ratio!
- **Arrhenius Energetics:** On the $\log k$ vs $\frac{1}{T}$ plot, the slope is strictly $-\frac{E_a}{2.303R}$. Catalysts lower the activation energy equally for both forward and reverse reactions, increasing rate constants without altering $\Delta H, \Delta G$, or equilibrium constant $K_{\text{eq}}$.

## UNIT III MASTER KINETICS FORMULA & HIGH-YIELD CHEATSHEET

| Core Concept / Law | Mathematical Expression | Application & CUET Problem Strategy |
|---|---|---|
| Stoichiometric Rate | $\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = +\frac{1}{c}\frac{d[C]}{dt}$ | Divide appearance/disappearance rates by stoichiometric coefficients. |
| Rate Constant Units | $k = (\text{mol L}^{-1})^{1-n}\text{s}^{-1}$ | Zero order: mol $\text{L}^{-1}$ $\text{s}^{-1}$; 1st: $\text{s}^{-1}$; 2nd: L $\text{mol}^{-1}$ $\text{s}^{-1}$. |
| Zero Order Integrated | $[R] = [R]_0 - kt \mid t_{\frac{1}{2}} = \frac{[R]_0}{2k}$ | Linear $[R]$ vs $t$ plot with slope $-k$. Half-life is proportional to $[R]_0$. |
| First Order Integrated | $k = \left(\frac{2.303}{t}\right)\log\left(\frac{[R]_0}{[R]}\right)$ | Linear $\log[R]$ vs $t$ with slope $-\frac{k}{2.303}$. Concentration-independent $t_{\frac{1}{2}}$. |
| First Order Milestones | $t_{75\%} = 2t_{50\%} \mid t_{99.9\%} \approx 10t_{50\%}$ | Every additional half-life halves the remaining reactant quantity. |
| Gaseous Decomposition | $k = \left(\frac{2.303}{t}\right)\log\left(\frac{p_i}{2p_i - p_t}\right)$ | Applies to $A(g) \rightarrow B(g) + C(g)$ where partial pressure $p_A = 2p_i - p_t$. |
| Pseudo-First-Order | $\text{Rate} = k'[A][H_2O] = k[A]$ | Ester hydrolysis and cane sugar inversion ($k = \left(\frac{2.303}{t}\right)\log\left(\frac{r_0 - r_\infty}{r_t - r_\infty}\right)$). |
| Arrhenius Equation | $k = Ae^{-\frac{E_a}{RT}}$ | Fraction of molecules with $E \geq E_a$ is $e^{-\frac{E_a}{RT}}$. |
| Arrhenius Linear Form | $\log k = \log A - \frac{E_a}{2.303RT}$ | Plot $\log k$ vs $\frac{1}{T}$: $y$-intercept $= \log A$; Slope $= -\frac{E_a}{2.303R}$. |
| Two-Temperature Ratio | $\log\left(\frac{k_2}{k_1}\right) = \left(\frac{E_a}{2.303R}\right)\left[\frac{T_2 - T_1}{T_1 T_2}\right]$ | Rate doubles every 10 K rise if $E_a \approx 53\text{kJ mol}^{-1}$. |
| Collision Theory | $\text{Rate} = PZ_{AB}e^{-\frac{E_a}{RT}}$ | Requires both threshold kinetic energy and proper steric orientation $P$. |
| Catalyst Action | Lowers $E_a$ to $E_{a'}$ equally for fwd/rev | Accelerates rate; does not change $\Delta H, \Delta G$, or equilibrium constant $K_{\text{eq}}$. |

## TOP 7 HIGH-FREQUENCY CUET TRAPS & PITFALLS

1. **Rate vs Rate Constant Units:** The units of reaction rate are ALWAYS mol $\text{L}^{-1}$ $\text{s}^{-1}$ regardless of order. The units of rate constant $k$ vary with order: $(\text{mol L}^{-1})^{1-n}$ $\text{s}^{-1}$. For a first order reaction, $k$ is expressed in $\text{s}^{-1}$ and is completely independent of concentration units.
2. **Order vs Molecularity:** Order can be zero, fractional, negative, or an integer and is strictly experimentally determined. Molecularity can ONLY be a positive integer $(1, 2, 3)$ and applies strictly to elementary reactions; complex reactions have NO overall molecularity.
3. **Half-Life Concentration Dependence:** For a first order reaction, $t_{\frac{1}{2}} = \frac{0.693}{k}$ is independent of initial concentration $[R]_0$. For a zero order reaction, $t_{\frac{1}{2}} = \frac{[R]_0}{2k}$ is **directly proportional** to $[R]_0$. If doubling $[R]_0$ doubles the half-life, the reaction order is ZERO, not first!
4. **First Order Multipliers vs 100% Completion:** For first order reactions: $t_{75\%} = 2 \times t_{50\%}$, $t_{87.5\%} = 3 \times t_{50\%}$, and $t_{99.9\%} \approx 10 \times t_{50\%}$. A first order reaction theoretically requires infinite time for 100% completion, whereas a zero order reaction completes fully at $t_{100\%} = 2 \times t_{\frac{1}{2}}$.
5. **Gas-Phase Pressure Kinetics ($A(g) \rightarrow B(g) + C(g)$):** At time $t$, partial pressure $p_A = 2p_i - p_t$ (NOT $p_i - p_t$). The integrated rate law is $k = \left(\frac{2.303}{t}\right)\log\left(\frac{p_i}{2p_i - p_t}\right)$. Always derive $p_A$ from stoichiometry!
6. **Catalyst Thermodynamics:** A catalyst increases the rates of forward and reverse reactions equally by providing an alternate pathway of lower $E_a$. It has ZERO effect on thermodynamic parameters: $\Delta H$, $\Delta G$, and equilibrium constant $K_{\text{eq}}$ remain completely unchanged.
7. **Arrhenius Plot Slope:** For a plot of $\log k$ versus $\frac{1}{T}$, the slope is $-\frac{E_a}{2.303R}$. For a plot of $\ln k$ versus $\frac{1}{T}$, the slope is $-\frac{E_a}{R}$. Forgetting the factor 2.303 in base-10 logarithmic problems is the most common numerical error in CUET!

## Practice Questions (Exam Pattern) — Part 1 (Q1 to Q5)

**Q1.** For the reaction $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, if the rate of disappearance of $H_2$ is $6.0 \times 10^{-4}$ mol $L^{-1}$ $s^{-1}$, the rate of appearance of $NH_3$ is:

**(A)** $4.0 \times 10^{-4}$ mol $L^{-1}$ $s^{-1}$ **(B)** $6.0 \times 10^{-4}$ mol $L^{-1}$ $s^{-1}$
**(C)** $9.0 \times 10^{-4}$ mol $L^{-1}$ $s^{-1}$ **(D)** $2.0 \times 10^{-4}$ mol $L^{-1}$ $s^{-1}$

**Sol. Q1.** Rate $= -\left(\frac{1}{3}\right)\left(\frac{d[H_2]}{d}t\right) = +\left(\frac{1}{2}\right)\left(\frac{d[NH_3]}{d}t\right) \Longrightarrow \frac{d[NH_3]}{d}t = \left(\frac{2}{3}\right)(6.0 \times 10^{-4}) = 4.0 \times 10^{-4}$ mol $L^{-1}$ $s^{-1}$.

**Answer: ( A )**

**Q2.** The rate constant for a reaction is experimentally found to be $k = 3.5 \times 10^{-4}$ L $mol^{-1}$ $s^{-1}$. The overall order of this reaction is:

**(A)** Zero order **(B)** First order
**(C)** Second order **(D)** Third order

**Sol. Q2.** Units of $k = (\text{mol L}^{-1})^{1-n}$ $s^{-1}$. For L $mol^{-1}$ $s^{-1}$ $= (\text{mol L}^{-1})^{-1}$ $s^{-1}$, $1 - n = -1 \Longrightarrow n = 2$ (second order).

**Answer: ( C )**

**Q3.** For reaction $A + B \rightarrow C$, doubling $[A]$ quadruples the rate, while doubling $[B]$ doubles the rate. The overall rate law and total reaction order are:

**(A)** Rate $= k[A][B]$, Order $= 2$ **(B)** Rate $= k[A]^2[B]$, Order $= 3$
**(C)** Rate $= k[A][B]^2$, Order $= 3$ **(D)** Rate $= k[A]^2[B]^2$, Order $= 4$

**Sol. Q3.** Rate $\propto [A]^2$ $(2^2 = 4)$ and Rate $\propto [B]^1$ $(2^1 = 2)$. Thus, Rate $= k[A]^2[B]$ with overall order $n = 2 + 1 = 3$.

**Answer: ( B )**

**Q4.** Which of the following statements regarding the molecularity and order of a chemical reaction is strictly INCORRECT?

**(A)** Molecularity is applicable to complex multi-step reactions as a whole **(B)** Order can be zero, fractional, or negative, but molecularity is always an integer from 1 to 3
**(C)** Order is determined experimentally, whereas molecularity is deduced from an elementary step **(D)** For elementary reactions, the order is always equal to its molecularity

**Sol. Q4.** Molecularity has no meaning for a complex reaction as a whole; it is meaningful only for individual elementary single-step reactions.

**Answer: ( A )**

**Q5.** A zero order reaction $A \rightarrow B$ has rate constant $k = 0.050$ mol $L^{-1}$ $s^{-1}$. If initial concentration $[A]_0 = 0.50$ M, its half-life $(t_{(\frac{1}{2})})$ is:

**(A)** 10 s **(B)** 2.5 s
**(C)** 5.0 s **(D)** 0.50 s

**Sol. Q5.** For zero order: $t_{\frac{1}{2}} = \frac{[A]_0}{2k} = \frac{0.50}{2 \times 0.050} = 5.0$ s.

**Answer: ( C )**

## Practice Questions (Exam Pattern) — Part 2 (Q6 to Q10)

**Q6.** The half-life of a first order reaction is 69.3 minutes. The value of its rate constant ($k$) is:

**(A)** $1.0 \times 10^{-2}$ min$^{-1}$
**(B)** $1.0 \times 10^{-3}$ min$^{-1}$
**(C)** $6.93 \times 10^{-2}$ min$^{-1}$
**(D)** $0.693$ min$^{-1}$

**Sol. Q6.** For first order: $k = \frac{0.693}{t_{\frac{1}{2}}} = \frac{0.693}{69.3} = 0.010$ min$^{-1}$ $= 1.0 \times 10^{-2}$ min$^{-1}$.

**Answer: ( A )**

**Q7.** For a first order chemical reaction, the time required for 75% of the reactant to decompose ($t_{(75\%\}}$) is related to its half-life ($t_{(50\%\}}$) by:

**(A)** $t_{75\%} = 1.5 \times t_{50\%}$
**(B)** $t_{75\%} = 2.0 \times t_{50\%}$
**(C)** $t_{75\%} = 3.0 \times t_{50\%}$
**(D)** $t_{75\%} = 4.0 \times t_{50\%}$

**Sol. Q7.** 75% completion leaves $25\% = \left(\frac{1}{2}\right)^2$ reactant. Each half-life halves the concentration, so $t_{75\%} = 2 \times t_{50\%}$.

**Answer: ( B )**

**Q8.** A radioactive sample of Carbon-14 ($t_{\frac{1}{2}} = 5730$ years) undergoes first order decay. What fraction of the original activity remains after $17,190$ years?

**(A)** $\frac{1}{4}$ (25%)
**(B)** $\frac{1}{16}$ (6.25%)
**(C)** $\frac{1}{8}$ (12.5%)
**(D)** $\frac{1}{32}$ (3.125%)

**Sol. Q8.** Number of half-lives $n = \frac{17190}{5730} = 3$. Remaining fraction $= \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$ (12.5%).

**Answer: ( C )**

**Q9.** Acid-catalysed hydrolysis of ethyl acetate ($CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$) is classified as a pseudo-first-order reaction because:

**(A)** The acid catalyst completely inhibits the backward reaction
**(B)** The reaction proceeds via a free-radical chain mechanism
**(C)** Both ethyl acetate and water react in equimolar amounts
**(D)** Water is present in large stoichiometric excess, so $[H_2O]$ remains constant

**Sol. Q9.** Water is the solvent ($\approx 55.5$ M) present in huge excess. Its concentration remains constant, yielding Rate $= k[CH_3COOC_2H_5]$.

**Answer: ( D )**

**Q10.** During the acid-catalysed hydrolysis of cane sugar (sucrose), the optical rotation of the reaction mixture changes because:

**(A)** Dextrorotatory sucrose hydrolyses into equimolar D-glucose and D-fructose with net laevorotation
**(B)** Dextrorotatory sucrose converts exclusively into dextrorotatory glucose
**(C)** Water breaks the asymmetric carbon atoms into optically inactive fragments
**(D)** Sucrose undergoes rapid mutarotation into non-reducing polysaccharide

**Sol. Q10.** Sucrose ($+66.5^0$) produces D-glucose ($+52.5^0$) and D-fructose ($-92.4^0$). Fructose's laevorotation dominates, inverting net optical rotation from (+) to (−).

**Answer: ( A )**

## Practice Questions (Exam Pattern) — Part 3 (Q11 to Q15)

**Q11.** According to the Arrhenius equation ($\log k = \log A - \frac{E_a}{2.303RT}$), a plot of $\log k$ against $\frac{1}{T}$ yields a straight line whose slope equals:

**(A)** $-\frac{E_a}{R}$ **(B)** $-\frac{E_a}{2.303R}$
**(C)** $+\frac{E_a}{2.303R}$ **(D)** $-\frac{2.303R}{E_a}$

**Sol. Q11.** Comparing $\log k = \left(-\frac{E_a}{2.303}R\right)\left(\frac{1}{T}\right) + \log A$ with $y = mx + c$ gives Slope $= -\frac{E_a}{2.303R}$.

**Answer: ( B )**

**Q12.** If the rate of a chemical reaction doubles for every $10^0$C rise in temperature, how many times will the rate increase when temperature rises from $20^0$C to $50^0$C?

**(A)** 3 times **(B)** 6 times
**(C)** 8 times **(D)** 16 times

**Sol. Q12.** Temperature increase $\Delta T = 50 - 20 = 30^0$C, corresponding to $n = \frac{30}{10} = 3$ intervals. Rate increases by $2^n = 2^3 = 8$ times.

**Answer: ( C )**

**Q13.** Which of the following parameters of a chemical reaction is altered by the addition of a positive catalyst at a constant temperature?

**(A)** Enthalpy of reaction ($\Delta_r H$) **(B)** Gibbs free energy change ($\Delta_r G$)
**(C)** Equilibrium constant ($K_c$) **(D)** Activation energy ($E_a$)

**Sol. Q13.** A catalyst lowers the activation energy ($E_a$) by providing an alternate pathway, but leaves thermodynamic functions ($\Delta H, \Delta G, K_c$) unchanged.

**Answer: ( D )**

**Q14.** According to collision theory, why do all collisions between reactant molecules not lead to product formation?

**(A)** Collisions only occur between molecules of the same chemical species **(B)** Only collisions with energy *ge* threshold energy and proper steric orientation are effective
**(C)** The pre-exponential factor is always negative at room temperature **(D)** Molecules must undergo electrolytic ionization prior to contact

**Sol. Q14.** Effective collisions require overcoming both the energy barrier ($EgeE_T$) and the steric orientation barrier ($P$).

**Answer: ( B )**

**Q15.** For a gas-phase first order reaction $A(g) \rightarrow B(g) + C(g)$, if $p_i$ is initial pressure of $A$ and $p_t$ is total pressure at time $t$, the rate constant $k$ is:

**(A)** $k = \left(\frac{2.303}{t}\right)\log\left[\frac{p_i}{2p_i - p_t}\right]$ **(B)** $k = \left(\frac{2.303}{t}\right)\log\left[\frac{p_t}{2p_i - p_t}\right]$
**(C)** $k = \left(\frac{2.303}{t}\right)\log\left[\frac{2p_i - p_t}{p_i}\right]$ **(D)** $k = \left(\frac{2.303}{t}\right)\log\left[\frac{p_i}{2p_i - p_t}\right]$

**Sol. Q15.** At time $t$: $p_A = p_i - x$, $p_B = x$, $p_C = x$. Total $p_t = p_i + x \Longrightarrow x = p_t - p_i$. Hence $p_A = p_i - (p_t - p_i) = 2p_i - p_t$. Substituting into 1st order: $k = \left(\frac{2.303}{t}\right)\log\left[\frac{p_i}{2p_i - p_t}\right]$.

**Answer: ( D )**